

Resolution: Substitution

Video Lesson

Duration: 00:09:25

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Lesson Transcript

In this lecture, we will look at substitution and unification, which are key steps in resolution for first-order logic.

Resolution for Predicate Calculus

The resolution step is valid for predicate calculus, when two clauses contain complementary predicates. For example, clause C_1 may contain predicate $P(\cdot)$ and clause C_2 may contain predicate $\neg P(\cdot)$.

$$\begin{aligned}C_1 : & \quad P(x) \vee Q(x) \\C_2 : & \quad \neg P(f(y)) \vee R(y)\end{aligned}$$

We could substitute $f(y)$ for x (replace x with $f(y)$) in C_1 and then resolve to get
 $C_3 : Q(f(y)) \vee R(y)$

The resolution step is valid for first-order logic when two clauses, for example, Clause C_1 , and Clause C_2 contain complementary predicates or literals. For example, Clause C_1 may contain the predicate $P(x)$, and Clause 2 may contain the predicate $\neg P(x)$. But what if these two arguments are different and Clause 1 is $P(x)$ and Clause 2 is $\neg P(f(y))$?

In this case, we can substitute $f(y)$ for x , replacing x with $f(y)$ in Clause C_1 . So, the first clause becomes $P(f(y)) \vee Q(f(y))$. And then, we can eliminate the complementary predicates and retrieve the resulting expression or the clause, which is $Q(f(y)) \vee R(y)$.

Remaining Issues

Theorem proving steps:

- 1 Conversion of natural language sentences into first-order logic formulas
- 2 Conversion to standard form
- 3 Resolution

Remaining issue: how to substitute variables to resolve two clauses and generate a new clause $\Rightarrow$ do substitution and unification.

To review the theorem-proving steps, we can take a look at these three steps.

First, we discuss the conversion of natural language sentences into first-order logic formulas, which we covered.

And conversion to standard form. Again, this we have covered in the previous lectures.

Finally, what remains is resolution. But here, the key problem becomes how to substitute variables to resolve two clauses and generate a new clause, and to do this, we need substitution and unification.

Substitution

- A **substitution**^I is a finite set of the form $\{v_1/t_1, \dots, v_n/t_n\}$ where each v_i is a variable, each t_i is a term (constant, variable, or function of terms), and no two v_i are identical.
- v_i/t_i means, v_i is replaced with t_i .
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- The *empty substitution* $\epsilon = \{\}$ contains no elements.

Why is substitution important: assists in resolving two clauses by making the two clauses with different variables compatible.

A substitution is a finite set of the form $\{v_1 / t_1, v_2 / t_2 \dots v_n / t_n\}$, where each v_i is a variable, and t_i on the right-hand side of the slash, is a term. As we have seen at the very beginning of the topic of first-order logic, a term can be either a constant, a variable, or a function.

One important requirement here is that the same variable should not repeat within this set.

So, each of these v_i / t_i means v_i is replaced with t_i . And also, there is a special case, which is called the empty substitution. We write it as ϵ , and it is simply an empty set. Again, it is very important to use substitution because this allows us to resolve two clauses with different variables in the predicates.

Substitution Applied to a Formula

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- Let $\theta = \{v_1/t_1, \dots, v_n/t_n\}$ be a substitution and E be an expression. Then $E\theta$ is an expression obtained from E with replacing *simultaneously* each occurrence of variable v_i ($1 \leq i \leq n$) in E by the term t_i .
 - $E\theta$ is called an instance of E .

In the textbook, $E\theta$ is denoted $SUBST(\theta, E)$.

Let us say we have a substitution, a set of variables, and term pairs like that, and let us call this set θ . Then when we apply the substitution to an expression, then what we need to do is to find out all occurrences of $v, v_1, v_2, \dots, v_n$ and replace them with $t_1, t_2, \dots, t_n$ in that expression, E .

$E(\theta)$ is how we write, applying the substitution θ to the expression E , which is called an instance of E .

In Russell & Norvig, this is written as $SUBST(\theta, E)$, but we will use this shorthand notation $E(\theta)$ throughout this course.

Substitution Examples

- $\theta = \{x/a, y/f(b), z/c\}, E = P(x, y, z)$
• $E\theta = P(a, f(b), c)$
- $\theta = \{x/f(z), y/z\}, E = P(x, y)$
• $E\theta = P(f(z), z)$
- $\theta = \{x/Socrates\}, E = \neg MAN(x) \vee MORTAL(x)$
• $E\theta = \neg MAN(Socrates) \vee MORTAL(Socrates)$

Here are examples of substitutions θ applied to expression E . Here, θ includes $\{x / a, y / f(b), z / c\}$. So, we replace x with a , and this is the result. We replace y with $f(b)$, and this is the result, and we replace z with c , and this is the result. So, this, as we have seen earlier, is the notation for applying this substitution to the expression E .

In the second case, we have $\{x / f(z), y / z\}$. So, x is replaced with $f(z)$, and y is replaced with z .

Last, we have a single variable and term pair, $\{x / Socrates\}$, and the expression can be more than just a single predicate. It can be any arbitrary first-order logic formula. So, let us say the expression was $\neg MAN(x) \vee MORTAL(x)$. In this case, when we apply this substitution to E , then what we get is $\neg MAN(Socrates) \vee MORTAL(Socrates)$.

One interesting thing that we need to note here is that on the left-hand side is always a variable. On the right-hand side can be a constant or a function, or even another variable because constants, functions, and variables are all terms.

Composition of Substitutions

Let $\theta = \{x_1/t_1, \dots, x_n/t_n\}$ and $\lambda = \{y_1/u_1, \dots, y_m/u_m\}$ be substitutions. Then the **composition** of θ and λ , denoted $\theta \circ \lambda$ is the substitution obtained from the set $\{x_1/t_1\lambda, \dots, x_n/t_n\lambda, y_1/u_1, \dots, y_m/u_m\}$ by deleting any element $x_j/t_j\lambda$ such that $t_j\lambda = x_j$ (e.g. x_k/x_k is meaningless) and any element y_i/u_i such that $y_i \in \{x_1, \dots, x_n\}$ (because y_i is already covered by θ).

Using this concept of substitution, we eventually want to construct an algorithm for unification. To do that, we need to introduce this concept of composition of substitutions.

Suppose we have two substitutions, θ and λ . Then, we want to compose them into θ composed λ so that when we apply this resulting substitution, then it would have the same effect as first applying the θ and then applying λ .

Composition of Substitutions

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The first step is copying over the content of the first substitution θ , whereas we apply this second substitution λ to the term part of the first substitution. So, instead of x_1 / t_1 , we have $x_1 / t_1 \lambda$, ..., and then $x_n / t_n \lambda$. So, that is the first component.

Composition of Substitutions



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The second step is simple. We just copy over the content of λ into the second part of the composed substitution.

Composition of Substitutions

Let $\theta = \{x_1/t_1, \dots, x_n/t_n\}$ and $\lambda = \{y_1/u_1, \dots, y_m/u_m\}$ be substitutions. Then the **composition** of θ and λ , denoted $\theta \circ \lambda$ is the substitution obtained from the set $\{x_1/t_1\lambda, \dots, x_n/t_n\lambda, y_1/u_1, \dots, y_m/u_m\}$ by deleting any element $x_j/t_j\lambda$ such that $t_j\lambda = x_j$ (e.g. x_k/x_k is meaningless) and any element y_i/u_i such that $y_i \in \{x_1, \dots, x_n\}$ (because y_i is already covered by θ).

But we have to be careful because if this y_1 already appears somewhere here, so this could have been x_2 , for example, then by the time we want to replace x_2 with something else, then it is already gone through this substitution. So, we want to get rid of these cases. So, that falls under case two.

Another one is after we apply λ to the t_j part of the first substitution θ , this may turn out to be the same as x_j . So, if we get x_j/x_j or x_k/x_k as a result of the initial substitution, we also want to eliminate those variables.

Examples: Composition of Substitution

Given

$$\begin{aligned}\theta &= \{x_1/t_1, x_2/t_2\} = \{x/f(y), y/z\} \\ \lambda &= \{y_1/u_1, y_2/u_2, y_3/u_3\} = \{x/a, y/b, z/y\}\end{aligned}$$

$$\begin{aligned}\theta \circ \lambda &= \{x_1/t_1\lambda, x_2/t_2\lambda, y_1/u_1, y_2/u_2, y_3/u_3\} \\ &= \underbrace{\{x/f(y)\lambda, y/z\lambda\}}_{\theta} \underbrace{\{x/a, y/b, z/y\}}_{\lambda} \\ &= \{x/f(b), \underbrace{y/y}_{\text{identity appeared in } \theta}, x/a, y/b, z/y\} \\ &= \{x/f(b), z/y\}\end{aligned}$$

Let us try a simple example. The first substitution θ is $\{x / f(y), y / z\}$. And the second substitution λ is $\{x / a, y / b, z / y\}$. So, first, we copy over the first substitution and apply λ to the term part.

Then we get $x / f(y) \lambda, y / z \lambda$. And again, we copy over the entire content of the second substitution here. So, this is what we have. So, λ calls for the replacement of y with b . So, this term here would become $f(b)$. And then here, z applied with the second substitution λ would turn z into y . So, this becomes y . And because the resulting substitution is identity, we have to eliminate this.

Examples: Composition of Substitution

Given

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Second, these two copy over, but x already appeared as the variable in the first substitution, and y also appeared earlier in the first substitution. So, we also get rid of these. And finally, only the last one from this copy over would remain in the final composed substitution. So, when we compose these two, we would get $\{x / f(b), z / y\}$.